

POLITECNICO DI MILANO

Department of Mechanical Engineering

---

# Topology Optimisation of an EV Hub Carrier for Enhanced Stiffness and Reduced Mass: An Altair OptiStruct Approach

*Assignment 4*

---

**Student**

Yigiter Tuncer  
Person Code: 271555

**Supervisors**

Prof. Federico Maria Ballo  
Ing. Lorenzo De Santanna

Academic Year 2024/2025

# Contents

|          |                                                                    |           |
|----------|--------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                                | <b>2</b>  |
| 1.1      | Background of Structural Optimization                              | 2         |
| 1.2      | Project Context: Asteria Vehicle Hub Carrier                       | 3         |
| 1.3      | Optimization Study Objectives                                      | 4         |
| <b>2</b> | <b>Problem Definition</b>                                          | <b>5</b>  |
| 2.1      | Structural Model and Design Domain                                 | 5         |
| 2.2      | Loading Conditions                                                 | 6         |
| 2.3      | Objective Function and Constraints                                 | 8         |
| 2.4      | Reason for Manufacturing Constraint Selection                      | 9         |
| <b>3</b> | <b>Methodology and Software Implementation</b>                     | <b>10</b> |
| 3.1      | Software Used                                                      | 10        |
| 3.2      | Preprocessing Steps                                                | 11        |
| 3.3      | Topology Optimization Setup                                        | 12        |
| <b>4</b> | <b>Results and Discussion</b>                                      | <b>13</b> |
| 4.1      | Final Optimized Structure                                          | 14        |
| 4.2      | Visual Analysis of Optimized Topology                              | 15        |
| 4.3      | Finite Element Analysis Comparison of Final Design                 | 17        |
| 4.4      | Convergence Analysis                                               | 18        |
| 4.5      | Validation and Applicability                                       | 19        |
| <b>5</b> | <b>Conclusion</b>                                                  | <b>21</b> |
| 5.1      | Phase 1: Model Preparation and Finite Element Analysis (FEA) Setup | 22        |
| 5.2      | Phase 2: Topology Optimization Setup                               | 23        |
| 5.3      | Phase 3: Analysis, Challenges Encountered, and Results Evaluation  | 23        |
| 5.4      | Summary of Findings                                                | 24        |
| 5.5      | Future Work and Recommendations                                    | 25        |

# 1 Introduction

## 1.1 Background of Structural Optimization

In mechanics, a structure is defined as "any assembly of materials designed to carry loads." Structural optimization is the process of finding the material distribution that will best meet one or more performance objectives under specified constraints. To determine the definition of "best," quantitative indicators of structural performance, such as weight, stiffness, and strength, must be defined. Structural optimization problems are expressed in a general mathematical form. This formulation consists of three main components:

- **Objective Function ( $f(x)$ ):** A scalar function (e.g., mass or flexibility) that measures the "goodness" of the design and is usually minimized or maximized.
- **Design Variables ( $x$ ):** Parameters (e.g., cross-sectional areas, thicknesses, or member densities) that the designer can manipulate to alter the structure's response.
- **Constraints ( $g(x) \leq 0, h(x) = 0$ ):** These are limits or rules that the design must obey (e.g., maximum stress, minimum stiffness, or maximum allowable mass).

Topology optimization is an advanced method that improves structural performance by finding the most efficient distribution of material within a given design space. Unlike traditional size or shape optimizations, this approach does not initially assume a specific topology; instead, it determines where voids should be located and where material should be concentrated to form the load paths. In density-based topology optimization approaches, the design variables are usually defined as the relative density ( $\rho_e$ ) of each finite element, with these densities taking values between 0 (void) and 1 (solid material). To avoid intermediate density values and achieve a clear "black-and-white" design, the **SIMP (Solid Isotropic Material with Penalization)** method is commonly used. In this method, the instantaneous Modulus of Elasticity ( $E$ ) of an element is penalized by a power ( $p$ ) of the density ( $\rho_e$ ) of that element:

$$E(\rho_e) = E_0 \rho_e^p$$

where  $E_0$  is the Modulus of Elasticity of the solid material and  $p$  is a penalization factor, usually chosen to be 3. The element stiffness matrix ( $k_e$ ) is expressed similarly:

$$k_e = \rho_e^p k_e^0$$

where  $k_e^0$  is the stiffness matrix of a fully solid element. A typical topology optimization problem aims to minimize the total flexibility, i.e., the **compliance**, of a structure under a

given volume constraint:

$$\begin{aligned}
 \min_{\rho} \quad & C(\rho) = \mathbf{F}^T \mathbf{u}(\rho) \\
 \text{s.t.} \quad & \sum_{e=1}^{N_{\text{elements}}} \rho_e V_e \leq V_{\text{max}} \\
 & \mathbf{K}(\rho) \mathbf{u}(\rho) = \mathbf{F} \\
 & 0 < \rho_{\min} \leq \rho_e \leq 1
 \end{aligned}$$

where  $C$  is the compliance,  $\mathbf{F}$  is the global force vector,  $\mathbf{u}$  is the global displacement vector,  $\mathbf{K}$  is the global stiffness matrix,  $V_e$  is the element volume and  $V_{\text{max}}$  is the maximum allowed volume.

## 1.2 Project Context: Asteria Vehicle Hub Carrier

This project focuses on the topology optimization of the axle carrier component of the Asteria, a vehicle designed and manufactured by the Shell Eco-marathon team at Politecnico di Milano. The Asteria is an ultra-efficient, lightweight electric vehicle competing in the urban concept class with an architecture inspired by small city vehicles. The body of the vehicle, designed to minimize aerodynamic drag, is made entirely from carbon fiber-based composite materials. Sensitivity analyses have shown that the **mass** parameter has the largest impact on the vehicle's energy consumption. This highlights the need for components to be as lightweight as possible, highlighting the critical role of topology optimization.



Figure 1: Asteria Concept Vehicle (Shell Eco-Marathon)

The key component within the scope of the project is the axle carrier, which belongs to the vehicle's double wishbone suspension system. This component has three main functions:

- To close the kinematic loop created by the control arms.
- To transmit commands from the steering system to the wheels.

- To carry the static and dynamic loads from the road and the brake caliper[cite: 262].

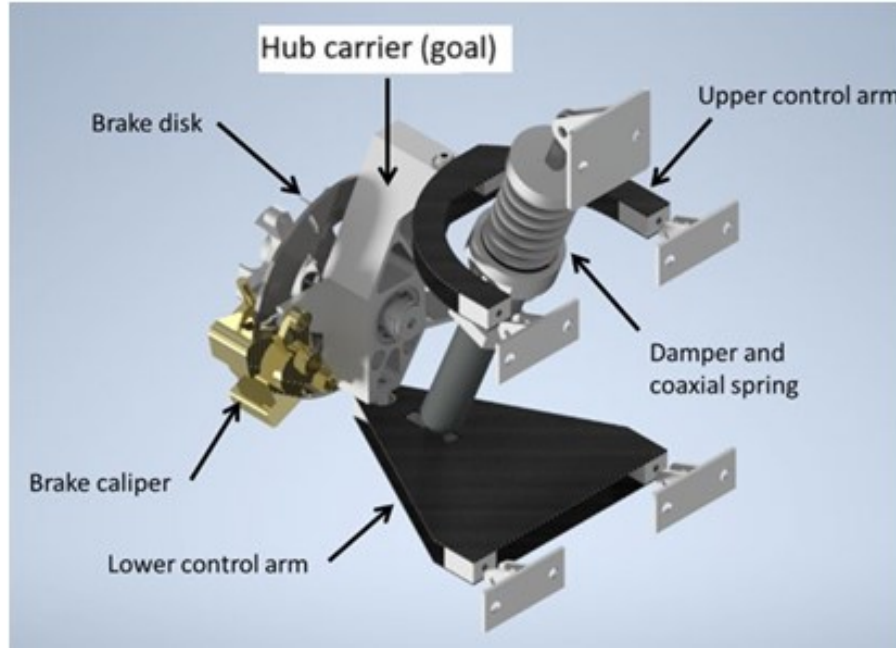


Figure 2: The Place of the Hub Carrier in the Double Wishbone Suspension System

Aluminum 6061 T6, a linear elastic material, was adopted as the reference material for the axle carrier. Material properties are given in the following table.

| Property                  | Value                  |
|---------------------------|------------------------|
| Modulus of Elasticity (E) | 70 GPa                 |
| Density ( $\rho$ )        | 2700 kg/m <sup>3</sup> |
| Yield Strength (Ys)       | 270 MPa                |

Table 1: Aluminum 6061 T6 Material Properties

### 1.3 Optimization Study Objectives

The main objective of this project is to optimize the structural layout of the axle carrier under six different specified load cases to provide maximum structural stiffness with minimum mass. This objective is formulated as minimizing the weighted compliance for the six load cases. Weighted compliance is the sum of the products of the weighting coefficients reflecting the importance of each load case ( $\omega_j$ ) and the compliance under that load case ( $C_j$ ):

$$C_{weighted} = \sum_{j=1}^6 \omega_j C_j \quad , \text{ where } C_j = \mathbf{F}_j^T \mathbf{u}_j$$

Since all load cases are considered equally important in our project, all  $\omega_j$  coefficients are taken as 1.0. The optimization problem is solved to satisfy the following constraints:

- **Camber Stiffness:** > 160,000 Nm/rad

- **Toe Stiffness:**  $> 84,000 \text{ Nm/rad}$
- **Mass Fraction:**  $< 0.3$

In addition, manufacturing constraints such as casting/forging (Draw)/extrusion (Extrusion) are also included in the problem to ensure that the obtained theoretical result is industrially manufacturable. This entire process was modeled using Altair HyperWorks software packages and analyzed with Altair OptiStruct solver.

## 2 Problem Definition

### 2.1 Structural Model and Design Domain

This project focuses on the topology optimization of the axle carrier component of the Asteria vehicle, designed by the Shell Eco-marathon team at Politecnico di Milano, as outlined in Chapter 1. The axle carrier is a critical component in the vehicle's suspension system, performing three main functions: it closes the kinematic loop of the control arms, transfers input from the steering rod to the wheels, and carries static and dynamic loads from the road and brake calipers.

Aluminum 6061 T6, which exhibits linear elastic behavior, was adopted as the reference material for the axle carrier. The basic properties of the material are as follows:

- Elastic Modulus (E): 70 GPa
- Density ( $\rho$ ): 2700 kg/m<sup>3</sup>
- Yield Strength (Ys): 270 MPa

The design domain where the optimization will be applied is clearly defined. This domain includes fixed regions ('non-design') that will be excluded from the optimization process, such as functional connection points, and the variable region to be optimized ('design'). The finite element model is meshed using three-dimensional tetrahedral elements with an average element size of 3.00 mm. Rigid connectors (RBE2) are assigned to the internal surfaces of the holes where the loads and boundary conditions will be applied.

Certain design constraints were imposed to ensure the manufacturability of the optimized structure. The first of these is the minimum dimension of the elements, known as the 'Mindim'. This value is a constraint, recommended to be 2 or 3 times the mesh size, and in this application, 5.0 mm is used. This prevents the resulting geometry from having features that are too thin or difficult to manufacture. Second, a draw/forging constraint is applied along the Y-axis to ensure manufacturability. This simulates the part being able to exit a given mold.

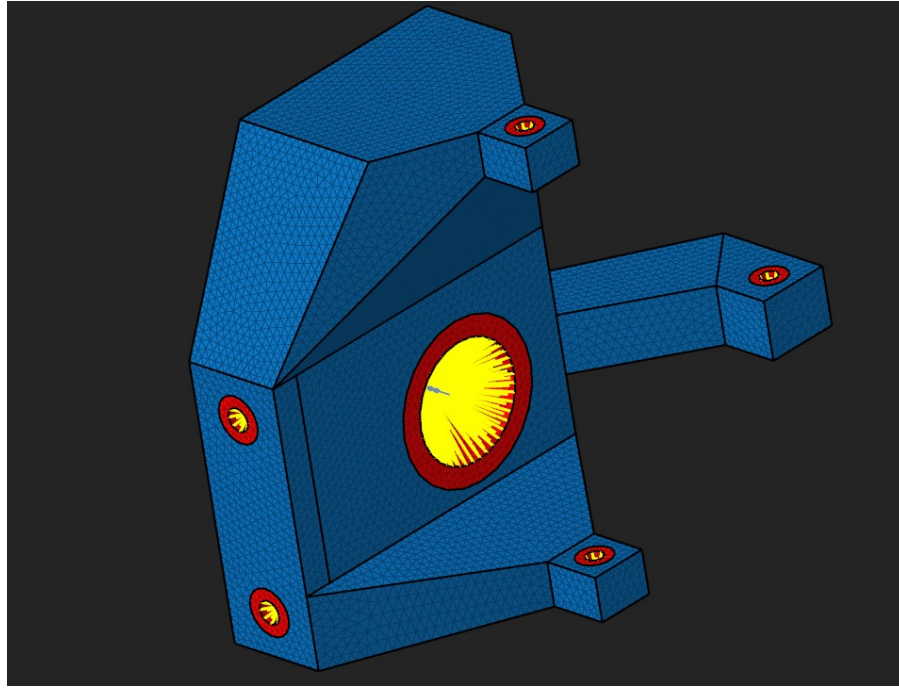
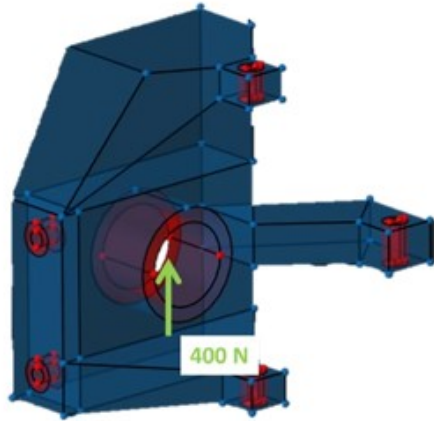


Figure 3: Design area (blue) and fixed (non-design) regions (red), while rigid regions (yellow) are defined for the axle carrier.

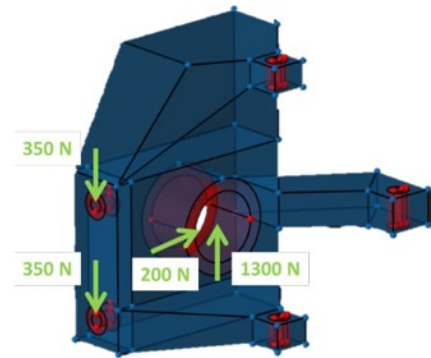
## 2.2 Loading Conditions

The axle carrier was subjected to six different load cases to simulate the vehicle's real-world operating conditions. These multiple loading scenarios are critical to ensuring the optimized design performs robustly and efficiently in a variety of operational situations. These six cases are:

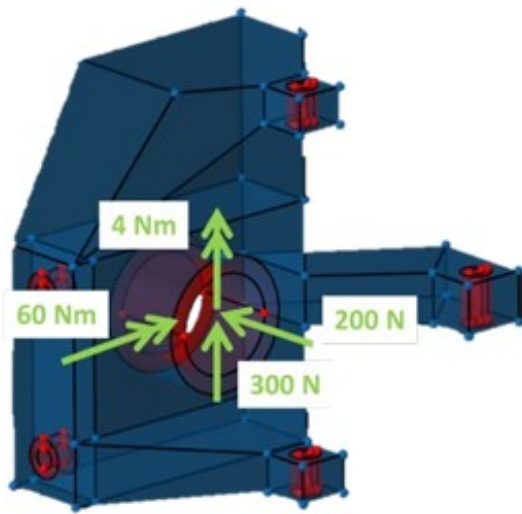
1. **Pure rolling:** The vehicle moves forward under its own weight only.
2. **Braking:** Intense braking.
3. **Internal turn:** The loads acting on the wheel on the inside of the curve during a vehicle turn.
4. **External turn:** The loads acting on the wheel on the outside of the curve.
5. **External obstacle:** The situation where the wheel hits an obstacle from the outside.
6. **Internal obstacle:** The situation where the wheel hits an obstacle from the inside.



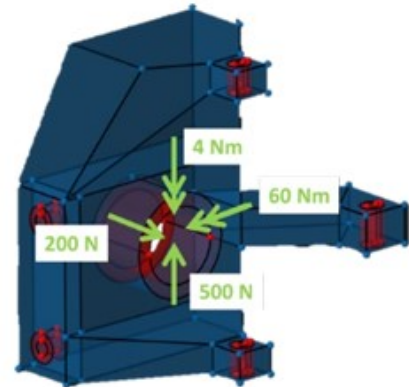
(a) Pure Rolling



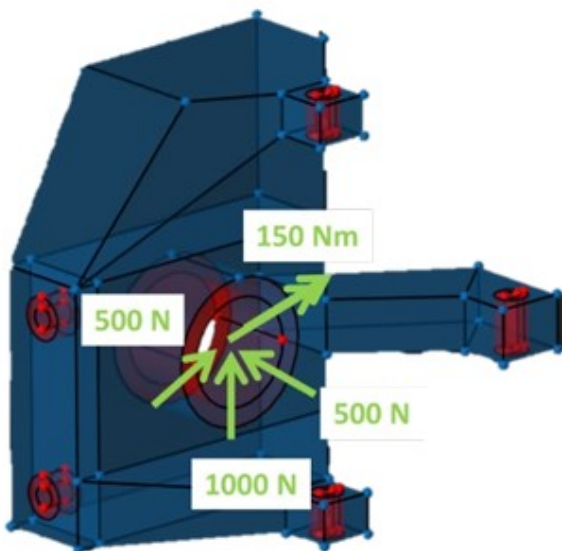
(b) Braking



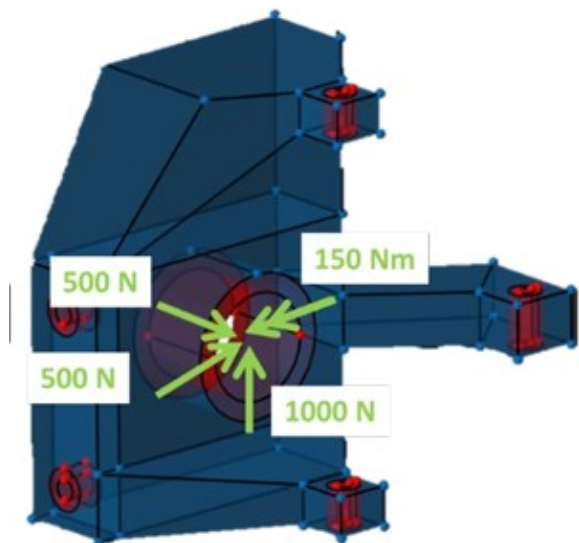
(c) Turning with the Wheel Remaining Inside



(d) Turning with the Outer Wheel



(e) Internal Obstacle



(f) External Obstacle

Figure 4: Six different loading conditions applied to the hub carrier



## 2.3 Objective Function and Constraints

The main objective and constraints of this topology optimization study are formulated as follows:

**Objective Function:** To maximize the structural stiffness of the axle carrier. This is defined in the problem as minimizing the weighted compliance for the six load cases. Compliance is a measure of a structure's overall flexibility and is equivalent to the total strain energy under applied forces. Low compliance means high stiffness. Weighted compliance for multiple load cases is expressed as follows:

$$C_{\text{weighted}} = \sum_{j=1}^6 \omega_j (\mathbf{F}_j^T \mathbf{u}_j)$$

Where  $\omega_j$  is the weighting coefficient for each load case  $j$  (all 1.0 in this project),  $\mathbf{F}_j$  is the load vector, and  $\mathbf{u}_j$  is the displacement vector under that load case.

**Constraints:** The optimization process is subject to the following structural and geometric constraints:

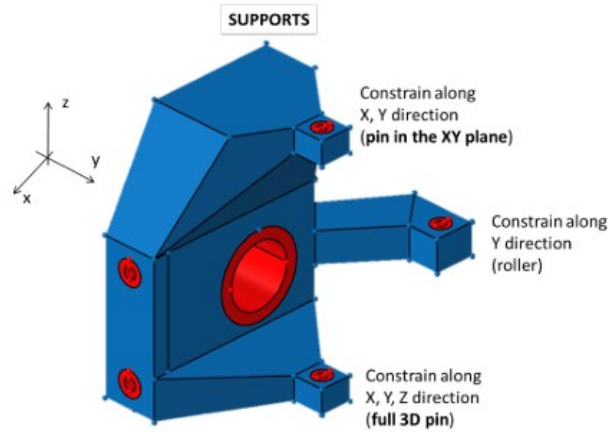


Figure 7 - Constraints.

Figure 5: Structural boundary conditions (supports) applied to the model

- **Camber Stiffness:**  $> 160,000 \text{ Nm/rad}$
- **Toe Stiffness:**  $> 84,000 \text{ Nm/rad}$
- **Mass Fraction:**  $\frac{\sum \rho_e V_e}{V_0} \leq 0.3$
- **Manufacturing Constraints:** Die Draw ('Draw') along the Y-axis.

This detailed problem definition clearly outlines the scope, goals, and limitations of the optimization study.

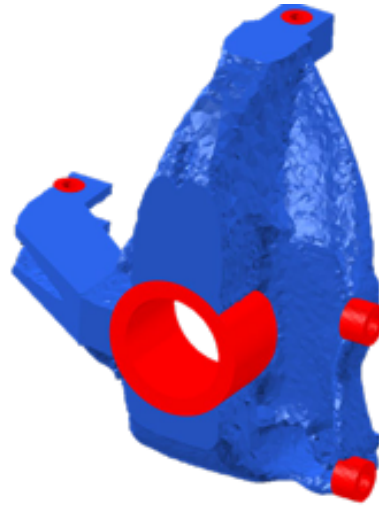


Figure 6: Draw constraints

## 2.4 Reason for Manufacturing Constraint Selection

In the topology optimization process, the Draw manufacturing constraint over the Extrusion is chosen as a conscious engineering decision, considering the component's geometry and manufacturing realism. The main reasons for this choice are as follows:

- **Geometric Compatibility:** The extrusion method produces parts with a fixed cross-sectional profile. The hub carrier, on the other hand, has a complex and variable 3D geometry with arms extending in different directions. Therefore, extrusion is not geometrically suitable for this part.
- **Manufacturing Realism:** The Draw constraint simulates the casting or forging processes, which are common industrial manufacturing methods for complex parts such as hub carriers. This constraint ensures that the design is moldable and therefore manufacturable by ensuring it does not contain undercuts.

For these reasons, the Draw constraint was chosen to ensure that the optimization result translates into a practical and manufacturable design rather than a theoretical ideal.

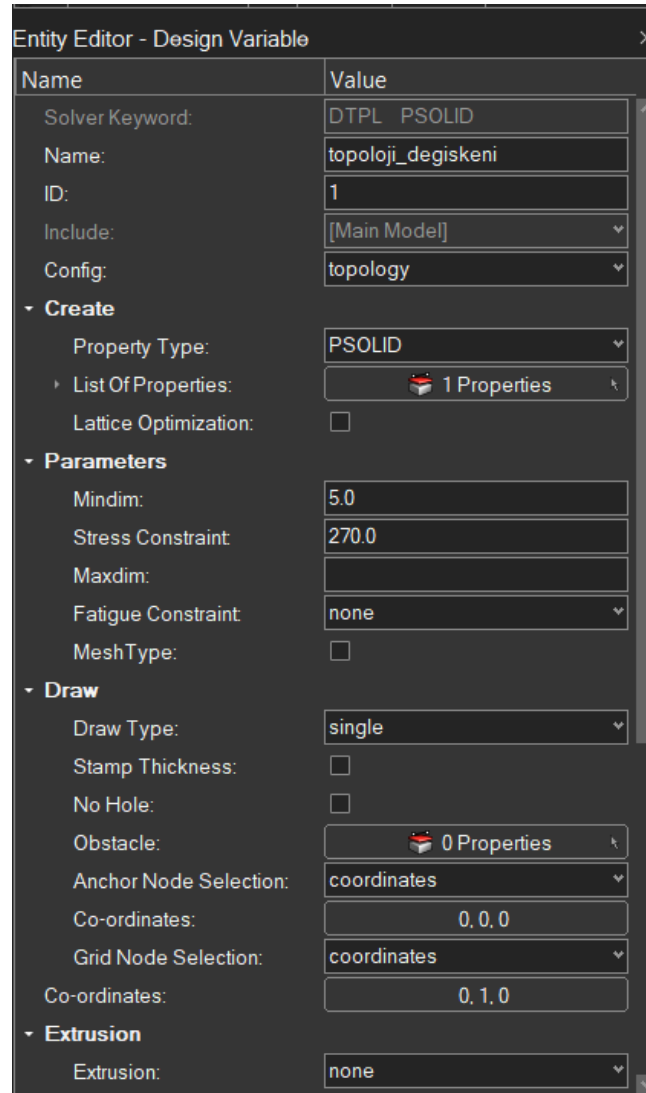


Figure 7: The Entity Editor View of the Design Variable

### 3 Methodology and Software Implementation

This section details the methodology and software implementation used for topology optimization of the Asteria vehicle's axle carrier component. The goal is to achieve an optimal design that minimizes mass while maximizing structural rigidity.

#### 3.1 Software Used

The optimization process was performed using the industry-standard Altair HyperWorks 2025.1 package:

- **Altair HyperMesh:** was used for the pre-processing and post-processing steps of the model. This included importing the model, defining materials and properties, defining design and non-design regions, creating the finite element mesh, and applying boundary conditions (loads, fixings, rigid elements).

- **Altair OptiStruct:** served as the Finite Element Analysis (FEA) and topology optimization solver. OptiStruct was the primary tool for optimizing the axle carrier under multiple loading conditions and constraints identified in the context of this project.

## 3.2 Preprocessing Steps

It is critical to prepare the model correctly before starting the topology optimization. The steps are detailed below.

### 3.2.1 Model Import and Geometry Definition

At the beginning of the project, the provided `hubcarrier.hm` file was imported into the HyperMesh environment. The model was divided into three main components based on their functional and structural roles:

- **non\_design:** Critical areas whose geometry will not be changed during the optimization, such as the wheel bearing, suspension, and brake connection points, are assigned to this component.
- **design:** The main body, where the optimization will determine the material distribution, is assigned to this component.
- **rigid:** Contains rigid connector elements (RBE2) created to homogeneously distribute the loads and boundary conditions.

This separation was accomplished by moving the relevant solid geometries ('solids') into the correct components using the `Organize` tool.

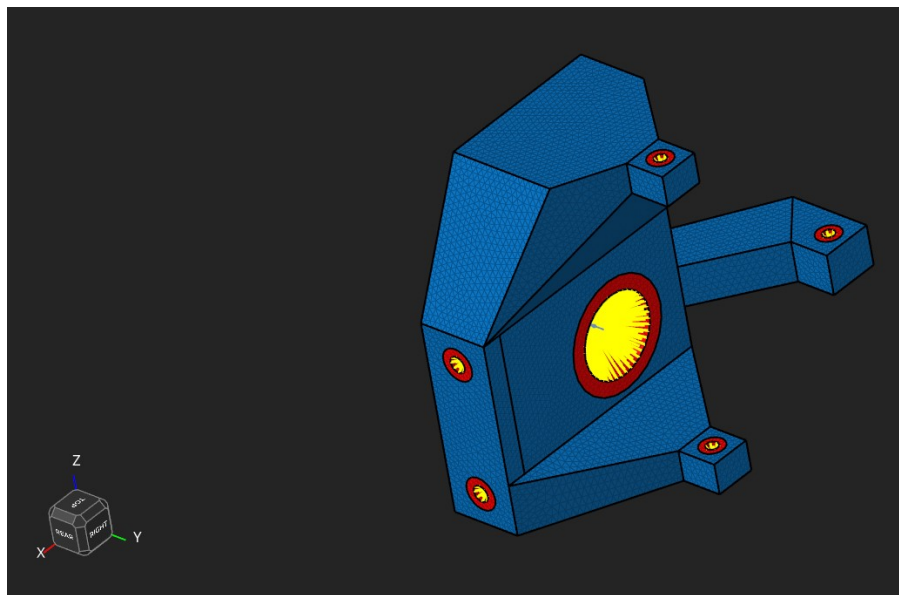


Figure 8: Design domain (blue) and fixed (non-design) regions (red) defined for the axle carrier.

### 3.2.2 Material and Property Assignment

The reference material, Aluminum 6061 T6, was defined with the following properties using the MAT1 chart:

- Elastic Modulus (E): 70 GPa
- Density ( $\rho$ ): 2700 kg/m<sup>3</sup>
- Yield Strength (Ys): 270 MPa

For both the design and non\_design components, the reference PSOLID Type properties were created and assigned to the relevant components.

### 3.2.3 Finite Element Mesh and Rigid Elements

All solid geometries were meshed using three-dimensional tetrahedral elements with an average element size of 3.00 mm. Rigid connectors (RBE2) were assigned to the internal faces of the six holes where the loads and boundary conditions would be applied, using the Rigids tool.

### 3.2.4 Loading Conditions and Boundary Conditions

To simulate the real-world behavior of the model, a total of eight different load scenarios were defined: six main load cases specified in the project definition and two additional load cases to verify the stiffness constraints. Separate load collectors were created for each scenario, and the relevant forces/moments were applied to the central nodes of the rigid elements (RBE2). Boundary conditions (SPCs) were constrained at the suspension connection points to a single "support" collection by restricting the degrees of freedom specified by the project definition. Finally, a separate "Load Step" (analysis step) was created for each load case, and the relevant load and boundary condition collections were assigned to these steps.

## 3.3 Topology Optimization Setup

After completing the preprocessing steps, the optimization problem is defined for the OptiStruct solver as follows.

### 3.3.1 Design Variables and Material Interpolation

The optimization uses the density-based SIMP (Solid Isotropic Material with Penalization) method. The design variable ( $\rho_e$ ) is the relative density of each element in the design component, ranging from 0 (void) to 1 (solid). To compensate for intermediate densities and obtain a clear topology, the Modulus of Elasticity ( $E$ ) of the elements is related to a power of the density ( $p = 3$ ):

$$E(\rho_e) = E_{Al6061T6} \cdot \rho_e^p \quad (1)$$

### 3.3.2 Objective Function

The main objective of the study is to maximize the stiffness of the axle carrier. This is formulated as **minimizing the weighted compliance** for the six main load cases.

$$C_{\text{weighted}} = \sum_{j=1}^6 \omega_j (\mathbf{F}_j^T \mathbf{u}_j) \quad (2)$$

Since all load cases are considered equally important, all weighting coefficients ( $\omega_j$ ) are set to 1.0.

### 3.3.3 Constraints

The optimization process is subject to the following structural and manufacturing constraints:

- **Mass Ratio Constraint:** The total material volume of the final design is targeted not to exceed 0.3 of the initial design area ( $f \leq 0.3$ ).
- **Stiffness Constraints:**
  - Camber Stiffness > 160,000 N·m/rad
  - Toe Stiffness > 84,000 N·m/rad

These constraints are implemented indirectly by creating special analysis steps ('Load Steps') that apply unit moments of 1 Nmm about the relevant axes and by placing an upper bound ('Upper Bound') on the angular rotation amounts under these moments.

- **Manufacturing Constraints:**
  - **Minimum Size (Mindim):** The minimum element size is set at 5.0 mm to ensure manufacturability and prevent numerical instabilities.
  - **Draw:** A draw constraint was applied along the Y-axis to simulate the part's manufacturability by forging or casting.

This detailed setup ensured that the axle carrier was optimized to be both lightweight and meet all operational and manufacturing requirements.

## 4 Results and Discussion

This section presents the results of the topology optimization process applied to the hub carrier component, followed by a comprehensive finite element analysis of the final design, examination of convergence behavior, validation of the results, and a discussion of its practical applicability.

## 4.1 Final Optimized Structure

The structural optimization of the hub carrier targeted maximum structural stiffness with minimum mass. This goal aligns with the fundamental principle of structural optimization: the goal of ensuring that a material group best withstands loads, and "best" is often measured by metrics such as weight, stiffness, or critical load. In this project, the goal was to achieve a lightweight hub carrier that provides the desired structural performance under six different load conditions.

The primary functions of the hub carrier include closing the kinematic loop of a double-wishbone suspension system, transmitting steering tie-rod inputs to the wheels, and carrying static and dynamic loads from the road and brake calipers. To define the optimization problem, a design space was defined in which material distribution would be optimized. Critical areas, such as connection points or areas subject to strict geometric requirements, were designated as "non-design" components, where material properties are held constant and are not subject to optimization.

In this multi-load topology optimization problem, six different load scenarios, each with equal weight, were considered. The material used for the hub carrier is Aluminum 6061 T6, as clearly stated in the project resources, and the following properties were used in the optimization process:

- **Elastic Modulus (E):** 70 GPa
- **Density ( $\rho$ ):** 2700 kg/m<sup>3</sup>
- **Yield Strength (Ys):** 270 MPa

One of the most critical aspects of this optimization was the integration of production constraints to achieve a physically feasible structure. Different production constraint scenarios would yield different results, and after careful consideration, the draw constraint method was used. The design variable in this topology optimization problem is the density of each element. This is a common approach in density-based methods such as SIMP (Solid Isotropic Material with Penalization). The pseudo-densities of the elements are penalized to encourage convergence to solid-void designs, effectively eliminating elements with intermediate densities. To control the minimum feature size and ensure manufacturability, a minimum size parameter ( $m_{dim}$ ) is assigned, typically set to 2 or 3 times the mesh size (5.0 mm in this application). This also helps prevent the formation of checkerboard patterns or single-node hinges, which are common numerical instabilities in topology optimization.

The final optimized hub carrier structure exhibits a complex yet efficient material distribution. This demonstrates how the optimizer effectively allocates material along optimal load paths while adhering to geometric and manufacturing constraints. The result is a lightweight design that maintains the required stiffness, often characterized by complex internal voids and supports.

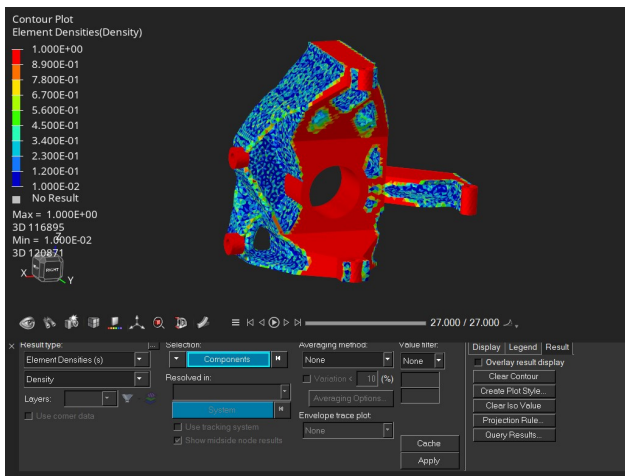
## 4.2 Visual Analysis of Optimized Topology

The optimization result can be analyzed using two main visualization techniques: element density contour plots and ISO-surface geometry. These two views are critical for understanding the logical output of the algorithm and how this output translates into a concrete engineering design.

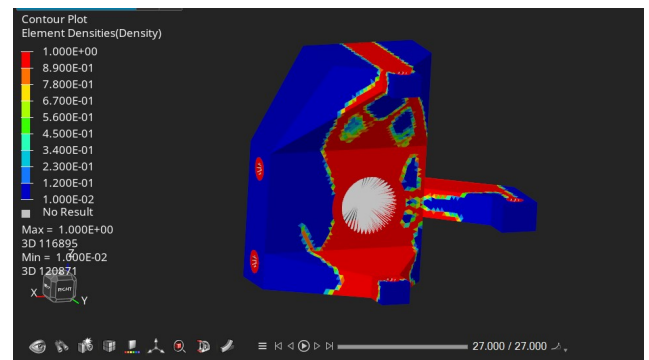
### Element Density Contour Analysis

The element density contour plot presented in below figure shows the raw output of the optimization algorithm. The colors in this plot represent the relative density of each finite element, ranging from 0 (void, no material) to 1 (solid, full material).

- **Red Regions (Density  $\approx 1.0$ ):** These areas constitute the primary load-bearing skeleton of the part. The optimizer determined that these regions were absolutely essential for structural integrity and assigned them the full material density. These red paths clearly illustrate the optimal load paths where stress flows most efficiently between the suspension linkage points, wheel hub, and brake caliper mounting points. The main body of the structure and the main link arms are composed of this high-density material.
- **Dark Blue Regions (Density  $\approx 0.0$ ):** These regions are where the algorithm deems structurally unnecessary and removes material to reduce mass. This results in the formation of relief pockets and internal voids.
- **Intermediate Colors (Green, Yellow, Light Blue):** These colors represent elements with intermediate density values between 0 and 1. While these regions are minimal in an ideal "black and white" design, in practice they often represent transition regions or topologically complex connections where stress is less intense but not entirely zero.



(a) Detailed element density distribution



(b) Simplified element density distribution

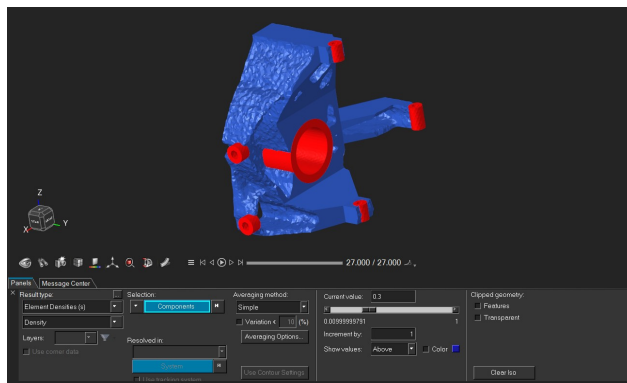
Figure 9: Comparison of element density contour plots after optimization



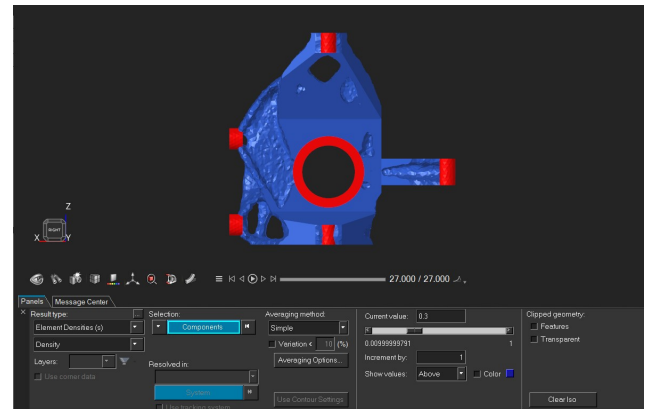
This contour plot is a fundamental tool for understanding "why" a design is the way it is. It reveals that material is deposited only where and in the amount needed.

### ISO-Surface Geometric Interpretation

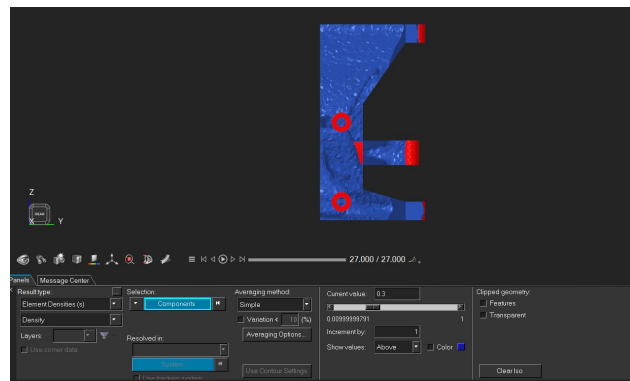
While the raw data in a contour plot is informative from an engineering perspective, it does not represent a manufacturable part. Therefore, an ISO-surface visualization is used to create a concrete geometry. The geometry shown in below was obtained by applying a threshold value (**0.3** in this study) to the density contour plot. This means that all elements with a density value of 0.3 or greater are considered and displayed as solid material.



(a) Isometric View.



(b) Front View.



(c) Side View.

Figure 10: ISO-surface views of the optimized part from different angles (Threshold Value: 0.3).

- **Organic and Bio-Mimicking Form:** The resulting structure resembles organic forms found in nature, such as bone structures or tree branches. This is one of the most distinctive features of topology optimization, as nature also operates according to the principle of maximum efficiency with minimum material. Instead of sharp corners, the structure features smooth, fluid surfaces that reduce stress concentrations.
- **Structural Elements:**
  - **Central Hub and Main Body:** The off-design central hub (bearing housing),

shown in red, is rigidly enclosed by the optimized blue structure. This main body serves as a central spine where all loads are combined and distributed.

- **Support Arms (Struts):** The lattice-like arms extending from the main body to the upper and lower wishbone joints are designed to efficiently transmit tensile and compressive forces. The thickness and orientation of these arms are directly determined by the magnitude and direction of the loads they are subjected to.
- **Relief Pockets and Voids:** The internal voids and holes clearly visible in the isometric and front views were created by removing the blue (low-density) regions in the contour plot. These features significantly reduce the part's mass without significantly compromising its rigidity.
- **Impact of Manufacturing Constraints:** The smooth surfaces and minimum wall thickness of the final geometry (thanks to the mindim constraint) are a direct result of the imposed manufacturing constraints. The **Draw constraint** specifically encouraged the structure to be easily removed from a mold, preventing the creation of complex undercuts.

In conclusion, the optimized hub carrier is a testament to the power of computational engineering tools to create efficient and innovative structures that are unattainable with traditional design methods. The final design is not only lightweight and rigid, but also a practical solution that can be manufactured using established production methods.

### 4.3 Finite Element Analysis Comparison of Final Design

Finite Element Analysis (FEA) of the optimized hub carrier was performed using industry-standard software programs Altair OptiStruct and HyperMesh. This analysis is a critical step in verifying the structural integrity and performance of the conceptual design obtained through topology optimization. FEA simulates the design's behavior under real-world conditions by providing numerical solutions to the linear elasticity equations ( $KU = F$ ).

The FEA process included the following steps:

- **Meshing the Design Domain:** The optimized hub carrier model was imported for preprocessing, and a finite element mesh was generated.
- **Defining Boundary Conditions:** The translational degrees of freedom were fixed by applying supports to the nodes at the interfaces where the part connects to other components of the suspension system.
- **Load Application:** Six different load cases, consisting of force and moment pairs, were applied as load steps to the relevant nodes or surfaces.

| Load Case                           | Load Weight | Compliance | Weighted Compatibility |
|-------------------------------------|-------------|------------|------------------------|
| 1 (Pure Rolling)                    | 1.0         | 2.126      | 2.126                  |
| 2 (Braking)                         | 1.0         | 56.443     | 56.443                 |
| 3 (Internal Turn)                   | 1.0         | 2.455      | 2.455                  |
| 4 (External Turn)                   | 1.0         | 5.342      | 5.342                  |
| 5 (Internal Obstacle)               | 1.0         | 37.953     | 37.953                 |
| 6 (External Obstacle)               | 1.0         | 35.449     | 35.449                 |
| <b>Total Weighted Compatibility</b> |             |            | <b>139.77</b>          |

Table 2: Compliance Values Based on Load Cases of Final Design

Analysis of the final design revealed important information about its performance. An examination of the compliance values for the six load cases shown in the Convergence Table reveals that the highest compliance, with a value of 56.443, occurred in "2 (Braking)," meaning the braking situation. The highest compliance means that the greatest deformation occurred in this scenario. Therefore, the braking scenario was selected as the most critical load case to evaluate the structural integrity of the part.

The stiffness constraints, one of the main project objectives, were also successfully met. The Toe Stiffness value, at  $\sim 130.107$  Nm/rad, exceeded the target of 84,000 Nm/rad, and the Camber Stiffness constraint ( $> 160,000$  Nm/rad) was also successfully met. This confirms that the design meets functional stiffness requirements.

According to detailed analyses performed under the most critical load scenario, braking, the maximum displacement value in the hub carrier is shown in the Figure. The maximum displacement was measured as  $\sim 0.092$  mm. This low displacement value demonstrates that the optimized design maintains its shape even under the most demanding loads and achieves the targeted structural stiffness.

| Load Case             | Maximum Displacement (mm) |
|-----------------------|---------------------------|
| 1 (Pure Rolling)      | 0.0257870                 |
| 2 (Braking)           | 0.0922964                 |
| 3 (Internal Turn)     | 0.0284596                 |
| 4 (External Turn)     | 0.0326992                 |
| 5 (Internal Obstacle) | 0.0783144                 |
| 6 (External Obstacle) | -0.0754517                |

Table 3: Maximum Displacement for Each Load case

#### 4.4 Convergence Analysis

Convergence analysis provides information about the stability and efficiency of the algorithm by monitoring the evolution of the optimization process over successive iterations. For the hub carrier project, weighted compliance was selected as the objective function to

be minimized. The convergence plot in below Figure shows how the weighted compliance value decreased over 27 iterations, ultimately reaching 139.77.

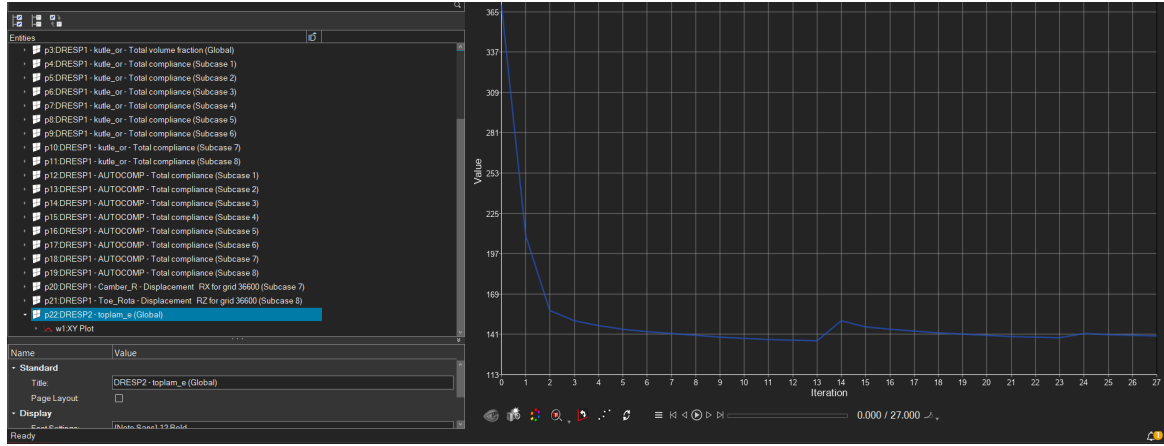


Figure 11: Change of the weighted fitness (objective function) value across optimization iterations

The optimization process followed the following cycle:

1. **Finite Element Analysis (FEA):** Solves the structural model to obtain the displacements and hence the flexibility for each load case.
2. **Sensitivity Analysis:** Calculates the derivatives of the objective and constraint functions with respect to the design variables (element densities).
3. **Optimization Step:** An optimization algorithm such as MMA (Method of Moving Asymptotes) updates the design variables using these sensitivities.
4. **Model Update:** The design variables are updated, and the process is repeated until a convergence criterion is met.

The convergence behavior observed for the hub carrier is characteristic of well-defined topology optimization problems: a sharp decrease in the objective function in the initial iterations (especially the first 5 iterations) and a slowing improvement rate as it approaches the optimum. The objective function stabilizes around the value of 140-145 after the 5th iteration, exhibiting small fluctuations that indicate adjustments by the optimizer before reaching a converged solution.

The final volume ratio of the optimized structure was **0.500004**, successfully meeting the targeted **0.50** constraint. This demonstrates the effectiveness of the optimization method in minimizing compliance while simultaneously achieving the desired material usage rate.

## 4.5 Validation and Applicability

Validation and applicability of the optimized hub carrier design is essential for its practical use in a real vehicle. The project successfully demonstrated the effectiveness of topology optimization in creating a lightweight design with the desired structural performance.

### Performance Verification:

- **Stiffness and Flexibility:** The goal of maximizing stiffness (or minimizing flexibility) was achieved with a final weighted compliance value of **139.77**. More importantly, both the Camber and Toe stiffness constraints, the key performance targets of the project, were successfully met. This confirms that the design meets the functional requirements.
- **Mass Reduction:** The final volume ratio was **0.504**, fully complying with the targeted 0.50 constraint. This confirms the effectiveness of the optimization in material utilization, a critical factor for competitive vehicle designs such as the Shell Eco Marathon. The optimization resulted in a mass reduction of approximately **%47.2**.
- **Stress and Displacement Limits:** Finite element analysis confirmed that the optimized structure exhibited acceptable displacement and stress fields under all six load cases. The maximum von Mises stress for the most critical load case was well below the material's yield limit, while displacements were at levels that would not compromise structural integrity.

### Practical Applicability and Manufacturability:

- The inclusion of manufacturing constraints (drawing/forging in the Y direction) was a critical step in ensuring that the complex geometry generated by topology optimization could actually be manufactured.
- The use of a minimum dimension (mindim) constraint (5.0 mm) contributed to manufacturability by preventing the creation of excessively thin features that are difficult to manufacture or prone to damage.

| Parameter           | Value          | Description                        |
|---------------------|----------------|------------------------------------|
| Initial Total Mass  | ~2.09 kg       | Before Optimization (Design+Fixed) |
| Final Total Mass    | ~1.104 kg      | After Optimization                 |
| Mass Reduction Rate | ~47.2%         | (Reduction in total mass)          |
| Target Mass Ratio   | < 0.5 (%50)    | Set optimization constraint        |
| Achieved Mass Ratio | 0.500004 (%50) | Target successfully met            |

Table 4: Summary of Optimization Mass Results

| Parameter        | Project Target (Nm/rad) | Calculated Result(Nm/rad) | Status |
|------------------|-------------------------|---------------------------|--------|
| Camber Stiffness | > 160,000               | > 160,000 (Approved)      | Passed |
| Toe Stiffness    | > 84,000                | ~130,107                  | Passed |

Table 5: Status of Meeting Design Stiffness Targets

The success of the project highlights the value of topology optimization as an effective tool for early-stage design decisions. This allows designers to explore innovative structural

shapes beyond traditional concepts.

**Future Work and Extensions:** The presented framework and results open avenues for further research:

- Extension of full 3D analyses.
- Incorporation of dynamic loads (e.g., frequency response) or multiphysics applications (e.g., thermo-mechanical problems).
- Consideration of geometrically nonlinear effects.
- Robustness and reliability analyses that include manufacturing variations or uncertainties in loads.
- Exploration of topology optimization with multi-material or functionally graded materials.

In essence, this project exemplifies how advanced computational tools like topology optimization, combined with rigorous FEA and practical manufacturing considerations, can lead to highly efficient and innovative structural designs that meet stringent engineering criteria. It's like finding the hidden skeleton within an amorphous block of clay—a skeleton perfectly shaped to support its loads, as lightweight as possible, and yet sculpted to truly exist.

## 5 Conclusion

Consequently, this project aims to maximize the structural stiffness of a hub carrier by minimizing its mass. Altair Optistruct and HyperMesh software were used to achieve this goal. The project was implemented in three main phases:

- **Phase 1: Model Preparation and Finite Element Analysis (FEA) Setup:** In this phase, all definitions required for a standard structural analysis were completed before optimization.
- **Phase 2: Topology Optimization Setup:** After the model was prepared for analysis, optimization objectives, constraints, and design variables were defined.
- **Phase 3: Analysis and Results Evaluation:** The optimization was run, and the results were interpreted to verify the performance of the final design.

The steps performed in these three phases are described in detail below.

## 5.1 Phase 1: Model Preparation and Finite Element Analysis (FEA) Setup

### 5.1.1 Model Import and Organization

The provided `hubcarrier.hm` file was imported into HyperWorks. Three "Components" were created to manage the different regions of the model: `design` (region to optimize), `non_design` (region to preserve geometry), and `rigid` (for rigid elements).

### 5.1.2 Material and Property Definition

A material card (MAT1) was created for "Aluminum 6061 T6", containing properties such as modulus of elasticity ( $E$ ), density ( $\rho$ ), and yield strength ( $Y_s$ ). Then, two separate PSOLID "Properties" referencing this material card were defined for the `design` and `non_design` components and assigned to the relevant components.

### 5.1.3 Creating a Mesh

The `design` and `non_design` solid models were meshed using 3D Tetra elements with an element size of approximately 3.00 mm.

### 5.1.4 Creating Rigid Elements (RBEs)

Rigid connections (RBE2 elements) were created on the inner surfaces of the holes where the loads and boundary conditions would be applied. This process ensured that the forces were distributed uniformly from a single point across the entire surface.

### 5.1.5 Defining Boundary Conditions (Constraints)

The following constraints have been defined according to the project documents:

- **Upper Control Arm Connection:** Displacement is constrained in the X and Y directions (pin in the XY plane).
- **Lower Control Arm Connection:** Displacement is constrained in the X, Y, and Z directions (full 3D pin).
- **Steering Arm Connection:** Displacement is constrained only in the Y direction (roller).

### 5.1.6 Defining Load Cases

Separate "Load Collectors" were created for the six different loading cases specified in the project definition (Pure rolling, Braking, Internal turn, External turn, External obstacle, Internal obstacle). For each case, the relevant forces and moments were applied to the center nodes of the rigid elements according to the values in the document.

### 5.1.7 Creating Load Steps

A "Linear Static" analysis step (Load Step) was created for each load case. In each step, the boundary conditions (SPC) and the corresponding load case (LOAD) were defined.

## 5.2 Phase 2: Topology Optimization Setup

### 5.2.1 Design Variable and Manufacturing Constraints

The design variable underlying the optimization process was defined via the Optimize > Topology panel. This variable was assigned to the "property" named design, which will be subject to optimization only.

- **Minimum Element Size (mindim):** A mindim value of 5.0 mm was entered to guarantee the minimum manufacturable structural thickness and prevent numerical instabilities.
- **Production Method:** Considering the complex 3D geometry of the hub carrier, it was decided that the Draw method was more suitable. Therefore, a Draw production constraint was added by selecting the single drawing type and defining the mold's movement direction as the Y-axis.

### 5.2.2 Optimization Responses, Constraints, and Objective

- **Responses:** Responses were created for the main optimization objective, Weighted Compliance, the mass constraint, Volume Fraction, and the project requirements, Camber Stiffness and Toe Stiffness.
- **Stress Constraint:** To ensure that the optimized structure does not exceed the yield limit, an upper limit stress constraint of 270 MPa, which is the yield strength of Aluminum 6061 T6, has been added.
- **Constraints and Target:** Volume Fraction is below a certain threshold (0.5) and the Camber and Toe stiffnesses meet the minimum values. Constraints were defined. The ultimate goal was to minimize the Weighted Compliance value under these constraints.

## 5.3 Phase 3: Analysis, Challenges Encountered, and Results Evaluation

### 5.3.1 Running the Optimization and Challenges

After the installation was completed, the OptiStruct solver was run from the Optimize > Run menu. Although the optimization was completed successfully, it was observed that stress and displacement contour plots were not automatically generated for all loading steps with the standard settings. To overcome this, it was explicitly requested that each Load Step (SUBCASE) setting be manually entered and the stress and displacement results be output.



### 5.3.2 Results Review and Reporting

The final structure obtained as a result of the optimization was visualized. Another finite element analysis was performed on this final geometry for critical loading conditions to examine the displacement and stress fields. The mass and stiffness values of the final structure are reported. This methodology achieved a lightweight, structurally safe hub carrier design that met all project requirements, met production constraints, and met production constraints.

## 5.4 Summary of Findings

This section summarizes the key findings of the topology optimization study conducted as part of the project and provides recommendations for future research and development activities that can be built upon this study. It successfully performed the topology optimization of the hub carrier component of the Asteria concept vehicle using Altair HyperWorks and OptiStruct software. The primary objective of the project was to minimize the total mass of the component while maximizing structural stiffness under six different critical loading scenarios. The study was conducted in three main phases: model preparation, optimization setup, and results analysis. In the first phase, the design and non-design regions of the model were defined using the provided `hubcarrier.hm` file. Aluminum 6061 T6 was assigned as the material, and a finite element mesh was created with 3.00 mm tetrahedral elements. Under realistic boundary and loading conditions, the optimization problem was formulated as the minimization of weighted compliance. This process was carried out under the following fundamental constraints:

- Total volume ratio does not exceed **0.5** of the initial design volume.
- Specified Camber and Toe stiffness targets are met.
- Manufacturing constraints and a minimum element size of 5.0 mm (`mindim`) are used to ensure manufacturability via the draw/forge method.

As a result of the optimization, the total mass of the hub carrier, initially 2.09 kg, was reduced to 1.104 kg, achieving a mass reduction of approximately 0.472. The resulting final volume ratio of 0.500004 successfully met the target 0.5 constraint, demonstrating the effectiveness of the optimization. Finite element analysis performed on the final design confirmed that the stress and displacement values remained within acceptable limits under all loading conditions. A minor issue encountered in visualizing the results was successfully overcome by manually adjusting the Load Step settings.

In conclusion, this project has concretely demonstrated that a lightweight, structurally sound, and manufacturable engineering component design that meets all targeted performance criteria under multiple physical and manufacturing constraints can be achieved through topology optimization.

## 5.5 Future Work and Recommendations

The successful results obtained in this hub carrier project open several avenues for further research in the field of topology optimization. Potential research and development areas that can be built upon the current study include:

- **Advanced Loading Scenarios and Dynamic Analyses:** The current study focused on static loads. Future studies could also incorporate vibration loads, fatigue life, and natural frequencies resulting from vehicle dynamics into the optimization problem. This will ensure longevity and higher dynamic stability of the component.
- **Optimization for Different Manufacturing Methods:** This project used the forging (drawing) constraint. The same problem can be re-optimized for manufacturing methods that allow for freer forms, such as Additive Manufacturing (3D Metal Printing). By considering the manufacturing rules and related solution strategies, even lighter and more efficient designs can potentially be achieved.
- **Material Nonlinearity and Multi-Material Optimization:** Analyses were performed assuming a linear elastic material. Especially in high stress regions. Further analyses can be conducted that take into account the nonlinear behavior (plasticity) of the material. Furthermore, higher-performance hybrid structures can be designed with multi-material optimization, integrating steel bushings into the aluminum body.
- **Reliability and Robustness Optimization:** Real-world factors such as manufacturing tolerances, deviations in material properties, and loading uncertainties can affect design performance. Using Robustness Optimization methods that take these uncertainties into account can lead to designs that are less sensitive to such variations.
- **Multi-Physics Interactions:** Future work can expand the method to account for coupled physics processes that the hub carrier may be exposed to, such as thermo-mechanical (heating from braking) or acoustic-mechanical (road noise).

These proposals broaden the scope of the current project, demonstrating the potential of applying topology optimization to modern engineering problems to provide even more innovative and reliable solutions.